

Overview of Lasso regression

Subject: Lasso regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|_1$$

1.

$b_1 = \begin{cases} (1 - \lambda / R^2) b_{\text{OLS}} & \text{if } \lambda \leq R^2 \\ 0 & \text{if } \lambda > R^2 \end{cases}$. $\lambda \leq R^2, \lambda > R^2$

2.

Here β_0 is the constant coefficient, $\beta := (\beta_1, \beta_2, \dots, \beta_p)$ is the coefficient vector, and t is a prespecified free parameter that determines the degree of regularization. Letting X be the covariate matrix, so that $X_{ij} = (x_i)_j$ and x_i is the i th row of X , the expression can be written more compactly as

3.

The first constraint is the lasso constraint, while the second directly penalizes large changes with respect to the temporal or spatial structure, which forces the coefficients to vary smoothly to reflect the system's underlying logic. Clustered lasso is a generalization of fused lasso that identifies and groups relevant covariates based on their effects (coefficients). The basic idea is to penalize the differences between the coefficients so that nonzero ones cluster. This can be modeled using the following regularization:

4.

where q_i is specified below and the "prime" symbol stands for transpose. The first fraction represents relative accuracy, the second fraction relative simplicity, and λ balances between the two.

5.

An R^2 of 75% means that in-sample accuracy improves by 75% if the unrestricted OLS solutions are used instead of the hypothesized β_0 values. The individual contribution of deviating from each hypothesis can be computed with the $p \times p$ matrix

6.

$$\min_{\beta \in \mathbb{R}^p} \left\{ \|y - X\beta\|^2 + \lambda \|\beta\|_1 \right\}$$

7.

$$\hat{\beta}_j = \frac{\sum_{i=1}^N x_{ij} y_i}{\sum_{i=1}^N x_{ij}^2 + \lambda} \quad \text{for } j = 1, \dots, p$$

8.

$$\min_{\beta \in \mathbb{R}^p} \left\{ \sum_{j=1}^J \beta_j^2 + \lambda \sum_{j=1}^J \beta_j \right\}, \quad z_j = (z_j^2)^{1/2}$$

9.

Group lasso In 2006, Yuan and Lin introduced the group lasso to allow predefined groups of covariates to jointly be selected into or out of a model. This is useful in many settings, perhaps most obviously when a categorical variable is coded as a collection of binary covariates. In this case, group lasso can ensure that all the variables encoding the categorical covariate are included or excluded together. Another setting in which grouping is natural is in biological studies. Since genes and proteins often lie in known pathways, which pathways are related to an outcome may be more significant than whether individual genes are. The objective function for the group lasso is a natural generalization of the standard lasso objective

10.

Using $X^T X = I$ and the ridge regression formula: $\hat{\beta} = (X^T X + N \lambda I)^{-1} X^T y$, yields:

11.

Least squares Consider a sample consisting of N cases, each of which consists of p covariates and a single outcome. Let y_i be the outcome and $x_i := (x_{i1}, x_{i2}, \dots, x_{ip})^T$ be the covariate vector for the i th case. Then the objective of lasso is to solve:

12.

Then $\hat{\beta} = \hat{\beta}^* + \lambda \frac{\hat{\beta}^*}{\sqrt{1 + \lambda \sum_{j=1}^p \hat{\beta}_j^2}}$, which, when the covariates are orthogonal to each other, gives

13.

Selected applications LASSO has been applied in economics and finance, and was found to improve prediction and to select sometimes neglected variables, for example in corporate bankruptcy prediction literature, or high growth firms prediction.

14.

$$\min_{\beta \in \mathbb{R}^p} \left\{ (y - X\beta)'(y - X\beta) + 2\lambda \sum_{i=1}^p |\beta_i - \beta_{i0}| \right\} \quad \left\{ \frac{(y - X\beta)'(y - X\beta)}{(y - X\beta_0)'(y - X\beta_0)} \right\} + 2\lambda \sum_{i=1}^p \left| \frac{\beta_i - \beta_{i0}}{\beta_{i0}} \right| \quad \left\{ \frac{\beta_i - \beta_{i0}}{\beta_{i0}} \right\} \right\}$$

15.

$$\min_{\alpha, \beta} \sum_{i=1}^N f(x_i, y_i, \alpha, \beta) \quad \text{subject to} \quad \|\beta\|_1 \leq t \quad \left\{ \frac{1}{N} \sum_{i=1}^N f(x_i, y_i, \alpha, \beta) \right\} \quad \text{subject to} \quad \|\beta\|_1 \leq t$$

16.

In contrast, variables can be clustered into highly correlated groups, and then a single representative covariate can be extracted from each cluster. Algorithms exist that solve the fused lasso problem, and some generalizations of it. Algorithms can solve it exactly in a finite number of operations.

17.

$$X_{(n+p) \times p}^* = (I_p + \lambda^2 I_p)^{-1/2} (X \lambda^2 I_p + I_p)^{-1/2} X^T, \quad \left\{ X_{(n+p) \times p}^* \right\}^* = (1 + \lambda^2)^{-1/2} \left\{ \binom{X}{\lambda^2 I_p} \right\}^* \left\{ \lambda^2 I_p + I_p \right\}^{-1/2},$$

18.

In statistics and machine learning, lasso (least absolute shrinkage and selection operator; also Lasso, LASSO or L1 regularization) is a regression analysis method that performs both variable selection and regularization in order to enhance the prediction accuracy and interpretability of the resulting statistical model. The lasso method assumes that the coefficients of the linear model are sparse, meaning that few of them are non-zero. It was originally introduced in geophysics, and later by Robert Tibshirani, who coined the term. Lasso was originally formulated for linear regression models. This simple case reveals a substantial amount about the estimator. These include its relationship to ridge regression and best subset selection and the connections between lasso coefficient estimates and so-called soft thresholding. It also reveals that (like standard linear regression) the coefficient estimates do not need to be unique if covariates are collinear. Though originally defined for linear regression, lasso regularization is easily extended to other statistical models including generalized linear models, generalized estimating equations, proportional hazards models, and M-estimators. Lasso's ability to perform subset selection relies on the form of the constraint and has a variety of interpretations including in terms of geometry, Bayesian statistics and convex analysis. The LASSO is closely related to basis pursuit denoising.